

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Nuclear Physics **L-T-P:** 3-0-0 **Credits:** 3

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Static properties of nuclei, Dipole and Quadrupole moments, Stability line, Binding energy curve, Liquid drop model, Semi-empirical mass formula, Shell model. (14)
Nuclear decays: Radioactivity, alpha decay: Gamow theory, Continuous beta spectrum and elementary ideas of Fermi Theory, gamma decay and Weisskopf estimates. (14)
Nuclear Force: two-nucleon system, Deuteron, General features of nuclear reactions. (7)
Radiation detection: Interaction of Ionizing Radiations with Matter, Ionization detectors, scintillation detectors. (7)

Reference Books:

1. Introductory Nuclear Physics: K. S. Krane
2. Theoretical Nuclear Physics: Blatt and Weisskopf

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Static properties of nuclei, Dipole and Quadrupole moments, Stability line, Binding energy curve, Liquid drop model, Semi-empirical mass formula, Shell model.	14
2.	Nuclear decays: Radioactivity, alpha decay: Gamow theory, Continuous beta spectrum and elementary ideas of Fermi Theory, gamma decay and Weisskopf estimates.	14
3.	Nuclear Force: two-nucleon system, Deuteron, General features of nuclear reactions.	7
4.	Radiation detection: Interaction of Ionizing Radiations with Matter, Ionization detectors, scintillation detectors.	7
Total number of hours		28